

Pharmacology of NSAIDs, Part 2

- Aspirin and Other Salicylates
- Nonselective NSAIDs
- Selective COX-2 Inhibitor
- Acetaminophen

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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After completing the preparation materials, students should be able to:

1. **Aspirin structure-activity relationship and pharmacokinetics:** Explain the mechanism and relevance of aspirin's short half-life and conversion to salicylic acid.
2. **Mechanism, uses and effects:** Correlate aspirin's therapeutic uses and adverse effects with its unique mechanism of action.
3. **Adverse effects:** Explain why aspirin is not used in pediatric patients with viral illness.
4. **Salicylate toxicity:** Identify the symptoms and signs of acute and chronic salicylate toxicity when presented with a case vignette.
5. **Non-aspirin NSAIDs:** Compare the mechanisms, effects, therapeutic uses, adverse effects, and precautions of the nonselective NSAIDs and COX-2 selective drug.
6. **Acetaminophen pharmacology:** Differentiate acetaminophen from NSAIDs in terms of mechanism, therapeutic effects, and adverse effect profile.
7. **Acetaminophen toxicity:** Relate acetaminophen's metabolic pathway to the formation of its toxic metabolite and to the risk factors that increase the potential for acetaminophen hepatotoxicity.

Preparation Materials

Required

- Practice Questions (questions and case vignettes FOR learning)

Highly relevant optional materials:

- Video Lecture | Dr. Goldstein's Notes | Guided Reading Questions | ScholarRx Bricks
- Textbook resources with links are listed on the next slide and on the CPG

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links to resources

ScholarRx Brick: Neurology and Special Senses > Disorders of the Nervous System and Special Senses > Pain Mechanisms and Treatment > Diagnostic Approach to Disorders of the Nervous System and Special Senses

Title of Brick: Nonopioid Analgesics <https://exchange.scholarrx.com/brick/nonopioid-analgesics>

Title of Brick: Renal Effects of NSAIDs: <https://exchange.scholarrx.com/brick/renal-effects-of-nsaids>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36: Nonsteroidal Anti-inflammatory Drugs, Disease Modifying Antirheumatic Drugs, Nonopioid Analgesics & Drugs Used in Gout > The section on Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical/content.aspx?bookid=3382§ionid=281752931>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 18: The Eicosanoids: Prostaglandins, Thromboxanes, Leukotrienes, & Related Compounds

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281749744>

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 40: Anti-inflammatory, Antipyretic, and Analgesic Agent

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253330080&bookid=3222>

Access Medicine Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 36: NSAIDs, Acetaminophen & Drugs Used in Rheumatoid Arthritis & Gout

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285596163>

PK:

- Oral / rectal: Taken up by platelets in the portal circulation | Rapid and saturable metabolism to active salicylic acid by esterases in GI mucosa, liver, plasma, and red blood cells | Very short half-life, ~20 minutes | Salicylic acid is metabolized in the liver; saturable at therapeutic doses

Mechanism:

- **Irreversibly** inhibits COX-1 and COX-2 by acetylating a serine residue in the hydrophobic channel, creating steric hindrance. AA is prevented from entering the catalytic channel.
- **Antiplatelet action:** Irreversible inhibition of COX-1 mediated TXA2 synthesis in platelets – the effect lasts for the life of the platelet (~8 days).
- **Salicylic acid:** Inhibits COX-2 gene transcription / Weak reversible inhibitor of COXs / Does not inhibit platelet aggregation

Therapeutic uses:

- Analgesic, anti-inflammatory, and antipyretic, especially at higher doses. Salicylic acid contributes to these effects.
- Antiplatelet: Low-dose aspirin for prevention of CV thrombotic adverse events: myocardial infarction (MI), transient ischemic attack (TIA), and stroke:

Adverse effects

- Upper GI irritation, peptic ulcers, bleeding | AERD | hypersensitivity reactions | tinnitus at high doses | associated with Reye syndrome in pediatric patients with viral illness

Key points: **NON-ASPIRIN NSAIDs**

PK:

- Well absorbed | Highly bound to plasma proteins | Widely distributed, including to synovial fluid with repeated dosing | Low first-pass metabolism | Good oral bioavailability | Nearly all undergo biliary excretion and enterohepatic circulation | Renal excretion of metabolites | Half-lives vary, ranging from short-acting ($t_{1/2} < 6$ hours) to long-acting ($t_{1/2} \geq \sim 12$ hours).

Mechanisms

- **nsNSAIDs:** Bind reversibly to the active sites in COX-1 and COX-2 and prevent AA from accessing the active site.
- **COX-2-selective agents:** Rigid, bulky structure fits into the larger, more flexible pocket of COX-2 but not the narrower channel of COX-1 → bulky molecule creates steric hindrance that prevents arachidonic acid from entering the catalytic channel.

Therapeutic uses:

- Analgesic, anti-inflammatory, antipyretic clinical applications

Adverse effects:

- Following slides

Key points: NSAIDs Adverse Effects

ALERT: NSAIDs are associated with significant adverse effects.

*NSAIDs should be used at the **lowest effective dose** for the **shortest duration possible**.*

GI adverse events:

- Dyspepsia, irritation, peptic ulcer, GI bleeding: Inhibition of cytoprotective effects of constitutive COX-1-derived prostanoids
- Agents with higher COX-2 selectivity may have a relatively lower risk for GI adverse effects – dose-dependent effect.

Bleeding risk:

- nsNSAIDs reversibly inhibit platelet COX-1 and can prolong bleeding time.
- Bleeding risk is increased when aspirin and nonselective NSAIDs are used together or with other antiplatelet agents, anticoagulants, or fibrinolytics.

Thrombosis risk and cardiovascular (CV) adverse effects:

- All nsNSAIDs and COX-2 selective drugs can increase the risk of MI and stroke.
- The mechanism is thought to be related to inhibiting COX-2-derived endothelial PGI₂ synthesis allowing unopposed pro-platelet aggregation effects of TXA₂.

Key points: **More NSAIDs Adverse Effects**

Renal adverse effects: Heart failure and kidney disease patients are at highest risk of renal adverse effects.

PGE2 and PGI2 maintain renal blood flow and glomerular filtration pressure by autoregulatory vasodilation of the afferent renal arteriole and promote sodium and water excretion.

Acute kidney injury:

- NSAIDs block vasodilatory action on afferent arteriole → vasoconstriction → ↓GFR → can lead to acute kidney injury in susceptible patients.

Edema and increased blood pressure:

- ↓ Prostanoid synthesis leads to sodium and water reabsorption →
- worsening hypertension, peripheral edema, exacerbation of heart failure, and other effects

Hyperkalemia:

- Indirect effect of inhibiting sodium excretion (a lecture for another day)
- Clinical relevance: Hyperkalemia can cause potentially life-threatening arrhythmias
- Patients at highest risk: chronic kidney disease, heart failure, diabetes, volume depletion, concurrent drugs that increase serum K⁺ levels (ACE inhibitors, ARBs, diuretics)

Key points: **More NSAIDs Adverse Effects**

Non-allergic Asthma Exacerbated Respiratory Disease (AERD / NERD):

- Ingestion of aspirin or NSAID by individuals with the triad of asthma, chronic rhinosinusitis with nasal polyposis and acute upper and lower respiratory tract hyperreactivity (nasal congestion and bronchospasm)
- NSAIDs are contraindicated in patients with aspirin-sensitive asthma.
- Proposed mechanism: NSAIDs block the prostanoid synthesis pathway but not leukotriene synthesis (lipoxygenase pathway). COX-1 inhibition shifts AA to the lipoxygenase pathway → leukotriene synthesis → inflammatory mediators of asthma.

Other adverse effects:

- Headache, dizziness
- Rash
- Elevated serum transaminases; hepatitis (rare)
- Immune-mediated hypersensitivity reactions
- Heme abnormalities (rare) – hemolytic anemia, agranulocytosis, neutropenia, thrombocytopenia, aplastic anemia

Key points: **Special Populations**

Pregnancy:

NSAIDs should be avoided throughout pregnancy unless the benefits to the mother clearly outweigh the risks to the developing fetus.

- **Oligohydramnios:** Beginning at 20 weeks gestation. Amniotic fluid is formed from fetal urine. NSAIDs interfere with fetal renal function, which can reduce the volume of amniotic fluid, leading to fetal adverse fetal effects.
- **Premature closure of patent ductus arteriosus (PDA):** NSAIDs inhibit PGE₂, which maintains the PDA in the fetus / greatest risk during third trimester
- **Suppression of labor:** NSAIDs inhibit PGE₂ and PGF₂ α , which mediate cervical ripening and uterine contractions.

Lactation: In general, NSAID use is considered compatible with breast feeding.

Pediatric patients: ***Lowest effective dose for the shortest duration possible***

- Guideline recommendations are available. Ibuprofen is approved for pediatric patients of all ages.

Older adults: If NSAID is needed: ***Lowest effective dose for the shortest duration possible.***

- Use only when the benefit outweighs the risk of adverse effects.

Key points: ACETAMINOPHEN

PK:

- Rapidly absorbed from GI tract | Hepatic glucuronide and sulfate conjugation → CYPs (notably CYP2E1) → toxic NAPQI → glutathione conjugation to a nontoxic metabolite | Urine excretion as metabolites | half-life 1-3 h (adults)

PK-related toxicity

- **Toxicity ALERT:** Saturation of the conjugation pathways and depletion of reduced glutathione can lead to NAPQI-induced hepatotoxicity, including hepatic necrosis requiring liver transplant.
- **IV acetaminophen Caution:** Not more effective than oral; Avoid prescribing, preparation, and administration errors that could result in accidental overdose and death.

Mechanism: Peroxidase site of COX-1 and COX-2

- **Acetaminophen** apparently inhibits the COX pathways in the CNS but does not appear to bind to the active site of the COX enzymes. Nociceptive effects may also be due to interactions with the endogenous opioid, cannabinoid, and descending serotonergic inhibitory pathways.
- **Peroxide tone hypothesis:** Acetaminophen is very effective in the low peroxide environment of the CNS (brain and spinal cord). It has very little anti-inflammatory effect in the high peroxide environments at sites of inflammation.

Key points: **ACETAMINOPHEN**

Therapeutic uses: Analgesic / antipyretic

- Temporary relief of minor aches and pains and reduction of fever
 - ❖ Acetaminophen is not used for anti-inflammatory effects.
- Frequently combined with opioids for enhanced analgesia at lower doses of each.

Adverse effects:

- Acetaminophen has few significant adverse effects.
- Hepatotoxicity may result from intentional or unintentional overdose.
 - ❖ Caution: Many OTC products contain acetaminophen in combination with other medicines.
 - ❖ Unintentional overdose can occur in pediatric and adult individuals.

Special populations:

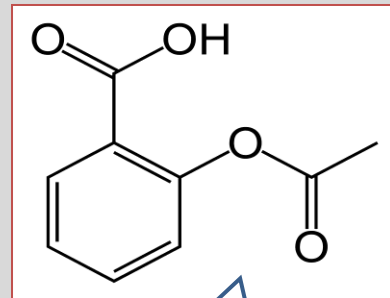
- Acetaminophen is considered safe for people of all ages and in pregnancy when used as directed.

Pharmacology of Salicylates

Aspirin

Non-aspirin (non-acetylated) salicylates

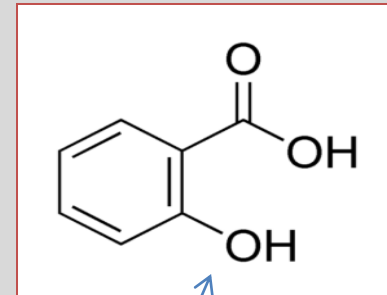
Aspirin



acetyl
group

acetylsalicylic acid: pKa 3.5

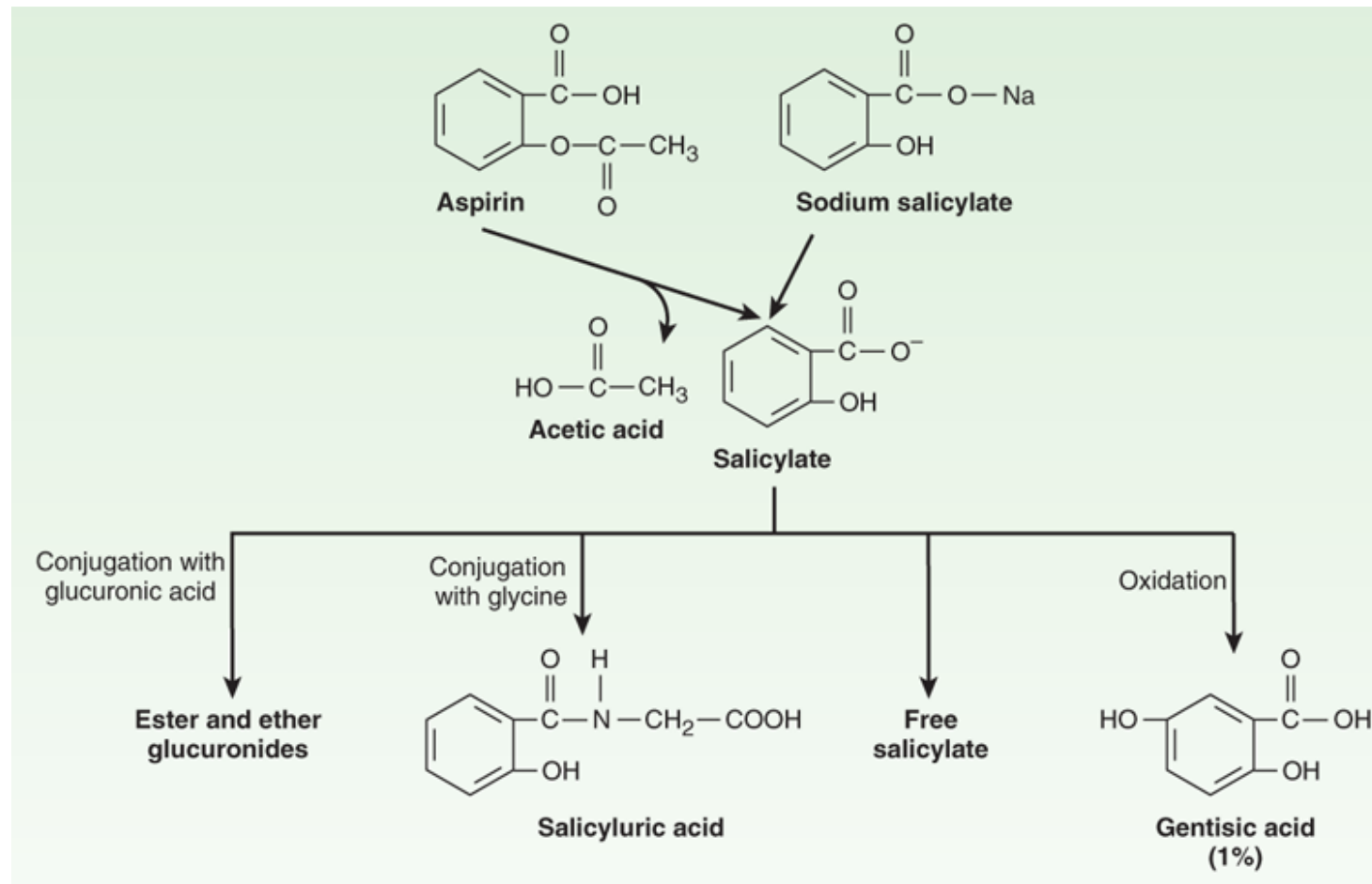
Salicylic acid



ortho position

Pharmacokinetics: Aspirin → Salicylic Acid

	Aspirin	Salicylic Acid
A	Oral: Rapid absorption mainly from upper small intestine Rectal: Slow, erratic absorption	Not administered systemically
D	Distributes into platelets in the portal circulation	Widely distributed Crosses BBB & placenta Salicylate protein binding (albumin) ~90% (saturable)
M	Esterases → salicylic acid (active) Esterase hydrolysis occurs in the GI mucosa, red blood cells, blood, liver, synovial fluid	Hepatic conjugation <i>next slide</i> Bioavailability of salicylic acid is 50%-75% Saturable metabolic processes
E	Unchanged aspirin reaches systemic circulation and is rapidly hydrolyzed.	Urine as salicyluric acid 75% salicylic acid 10% glucuronide conjugates
t _{1/2}	~20 minutes (due to rapid ester hydrolysis)	Low dose: 2-3 hours High dose: 10 hours } first order → zero order kinetics



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Structure and metabolism of the salicylates.

Reproduced with permission from Meyers FH, Jawetz E, Goldfien A: Review of Medical Pharmacology, 7th ed. New York, NY: McGraw Hill; 1980.

Aspirin Mechanism of Action

Irreversible acetylation of the cyclooxygenases

Aspirin covalently acetylates a serine residue within the active sites of COX-1 and COX-2.

COX-1 Ser530, COX-2 Ser516

Steric hindrance prevents AA from entering the catalytic channel of the enzyme.

AA: arachidonic acid

Prostanoid synthesis is prevented.

Salicylic acid mechanism:

In vitro studies suggest salicylic acid inhibits AA metabolism, at least in part, by inhibiting transcription binding to COX-2 promoter and mRNA induction → suppresses inflammatory upregulation of COX-2.

Salicylic acid is a weak reversible inhibitor of COX-1 / COX-2 and does not have antiplatelet effects.

Mechanism of aspirin's antiplatelet (cardioprotective) actions

Low-dose aspirin inhibits COX-1-derived platelet TXA2 synthesis mainly in the portal system.

1x daily dose

Rapid ester hydrolysis to salicylate in GI mucosa, liver, plasma, red blood cells, synovial fluid



short half-life

Oral low-dose aspirin enters platelets in the portal system



Irreversibly acetylates and inactivates platelet COX-1



Prevents TXA2 synthesis for the life of the platelet (8-10 days)

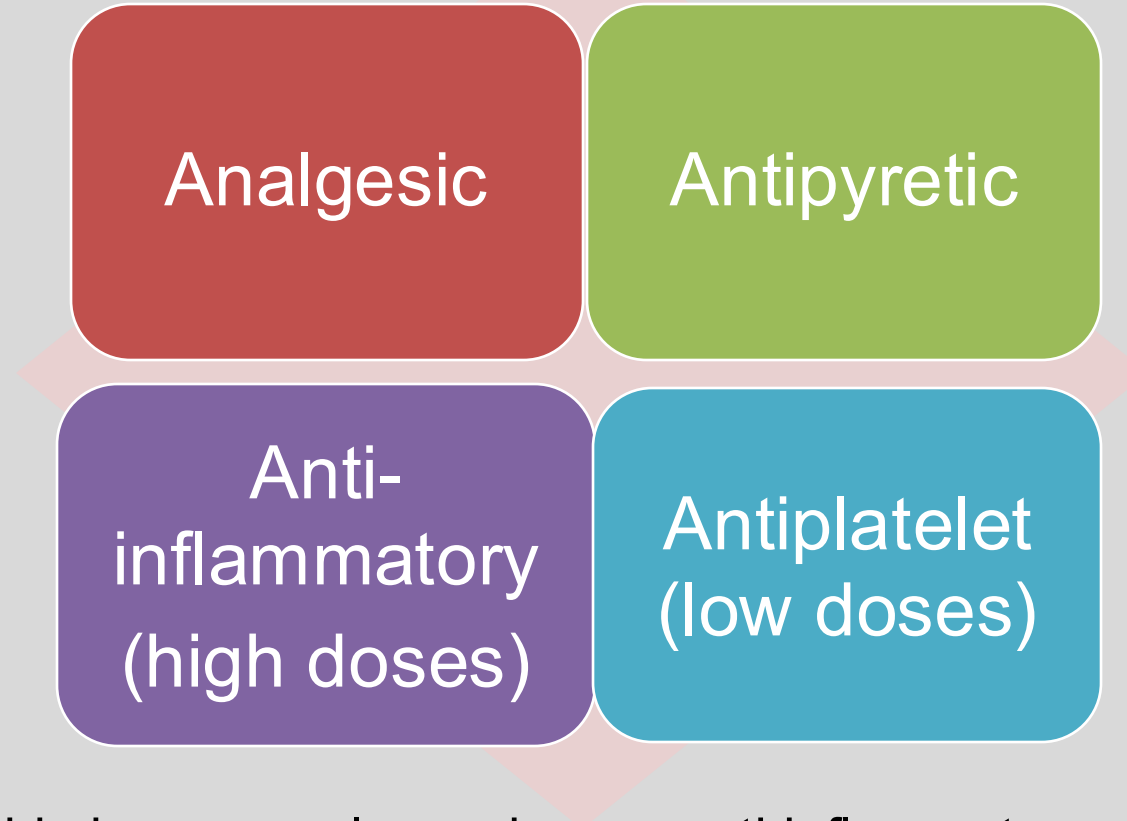


Inhibits platelet aggregation and local vasoconstriction

Low-dose aspirin does not inhibit endothelial COX-2 in the systemic vasculature

Low-dose aspirin use shifts vascular TXA2 – PGI2 balance toward PGI2 mediated effects: Inhibition of platelet aggregation, inducing platelet disaggregation, and vasodilation.

Aspirin's Effects and Therapeutic Uses



Aspirin is now rarely used as an anti-inflammatory agent.

- Much higher risk of GI toxicity than with other NSAIDs.
- Higher frequency of tinnitus, metabolic acidosis, renal dysfunction, and bleeding with the very high anti-inflammatory doses.

Therapeutic uses: Aspirin and Salicylic acid

Aspirin

- Pain: mild-moderate
- Fever (not for children/adolescents)
- Kawasaki disease (first line with IVIG)
- Rheumatic fever (alternative NSAID)
- Antiplatelet effects for secondary prevention of adverse cardiovascular events – myocardial infarction (heart attack) and stroke
- Niacin tolerability (inhibition of PGD2 synthesis reduces flushing, tingling)

IVIG: intravenous immune globulin

Salicylic acid: Topical uses

Salicylic acid: Keratolytic

(keratolytic: an agent that causes softening and shedding of the horny outer layer, keratin, of the skin)

Acne, calluses, corns, warts, dandruff, seborrheic dermatitis, hyperkeratotic skin disorders

Methylsalicylate:

Musculoskeletal pain

ointments, liniments, patches

Salicylic acid is absorbed through the skin but rarely reaches toxic levels when used as directed.

Aspirin: Dose-related adverse effects

Most common (can occur even at low doses):

Bleeding
disorders

Antiplatelet effect increases risk of bleed

Peptic ulcer

Local cellular injury *plus*

Erosive gastritis

Inhibition of gastric protective effects of prostanoids *and*
Increases risk of GI bleeding

Asthma

Increased risk of aspirin sensitivity

High doses:

Tinnitus, vertigo, hyperventilation, and respiratory alkalosis

Very high doses:

Metabolic acidosis, dehydration, tachypnea, hyperpnea, seizures, cardiovascular collapse, coma, and death

Aspirin's potentially serious adverse effects (occur rarely)

Reye syndrome:

- Aspirin use in **children and adolescents** with viral illness (chicken pox, influenza) was rarely associated with Reye syndrome
 - Rates of Reye syndrome fell dramatically after advisories (1980s) against aspirin use in febrile children.
- Severe and often fatal
- Characterized by the acute onset of encephalopathy, liver dysfunction, and fatty infiltration of the liver and other viscera

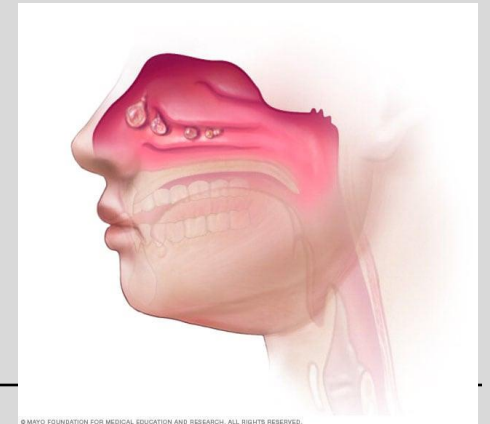
Aspirin-exacerbated respiratory disease (AERD) (“aspirin triad”)

Non-allergic reaction with ingestion of aspirin (or NSAID → NERD)

1. Bronchospasm / Asthma (upper and lower respiratory hyperreactivity)
2. Chronic rhinosinusitis
3. Nasal polyposis

Hepatotoxicity: Associated with high dose, chronic use; delayed onset; reversible when aspirin is withdrawn

mechanism, next slide



<https://www.mayoclinic.org/diseases-conditions/nasal-polyps/symptoms-causes/syc-20351888>

Aspirin-, NSAID-exacerbated respiratory disease (AERD / NERD)

Proposed mechanism

NSAID-mediated inhibition of COX-1 pathway may **shunt** arachidonic acid to the lipoxygenase pathway



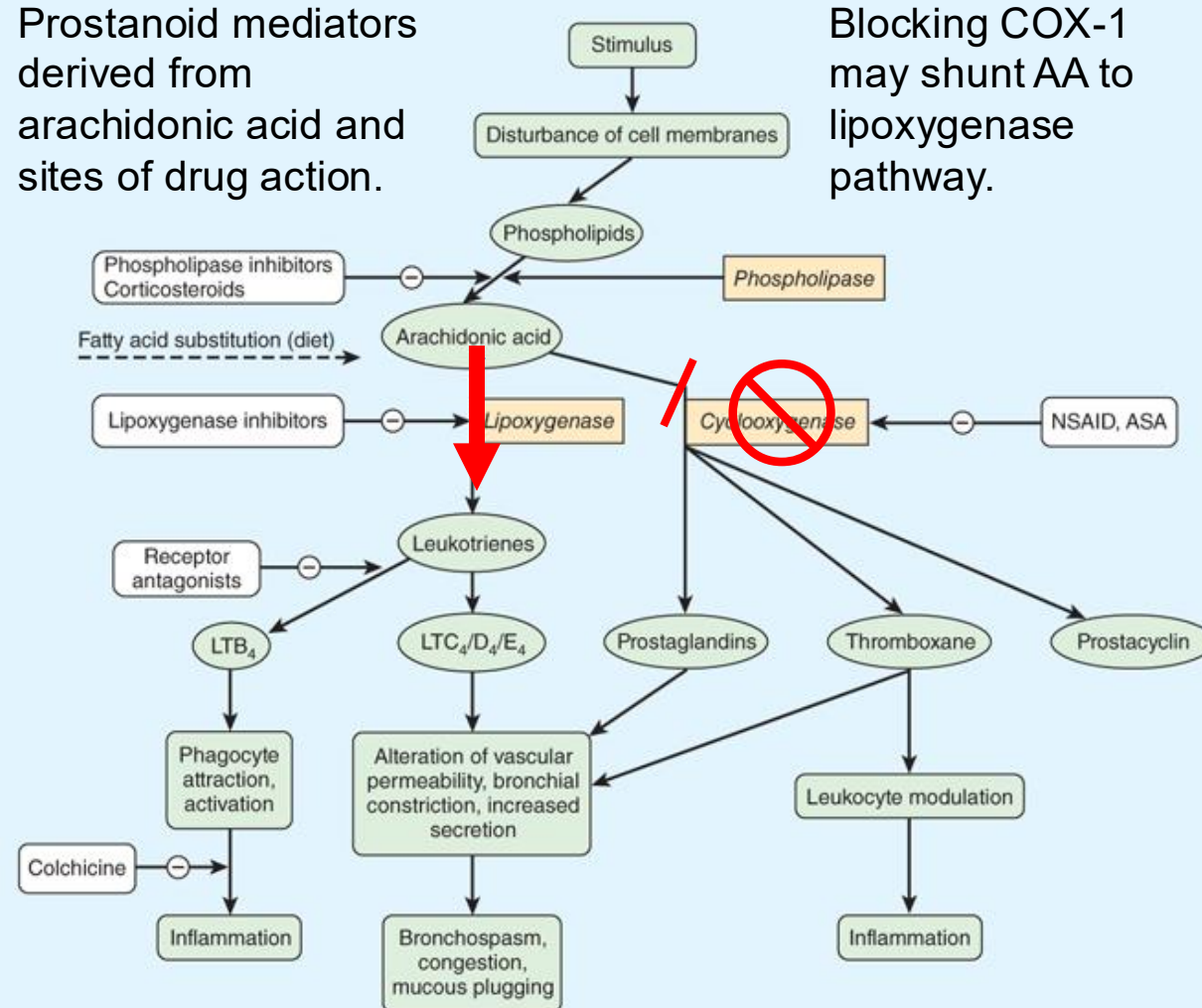
overproduction of leukotrienes



upper and lower airway inflammation
→ nasal congestion and bronchospasm

Prostanoid mediators derived from arachidonic acid and sites of drug action.

Blocking COX-1 may shunt AA to lipoxygenase pathway.



Source: Bertram G. Katzung, Anthony J. Trevor: Basic & Clinical Pharmacology, 13th Ed.
www.accesspharmacy.com
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ASA, acetylsalicylic acid (aspirin); LT, leukotriene; NSAID, nonsteroidal anti-inflammatory drug; Katzung Figure 36-2

Salicylate Toxicity (salicylism) (TAKEAWAY: See next slide.)

Acute toxicity: Unintentional or intentional ingestion of aspirin or co-ingestion of salicylate-containing products. Morbidity and mortality increase when the toxicity is not rapidly identified, if there is a delay in starting treatment, or if the poisoned patient is not treated aggressively.

Mild chronic toxicity: Daily higher doses (even within the therapeutic range) over time or co-ingestion of aspirin-containing products.

Clinical findings in chronic and acute salicylate poisoning overlap but symptoms of chronic toxicity can be milder, ***often go unrecognized or attributed to another condition.***

Presentation: Salicylate toxicity causes a variety of metabolic disorders.

Early symptoms: Nausea, vomiting, headache, dizziness, tinnitus (classic symptom)

Metabolic disorders
Early and later s/s
Acid/base disorder

More significant toxicity: Tachypnea (fast breathing), hyperpnea (deep breathing), hyperthermia, diaphoresis, dehydration, altered mental status (agitation → lethargy), noncardiac pulmonary edema, seizures, coma

Mechanism of acid/base disorder in salicylate toxicity:

Early salicylate toxicity → direct stimulation of respiratory center in medulla → respiratory alkalosis

Uncoupling of oxidative metabolism → $\uparrow\text{CO}_2$ → $\rightarrow\downarrow\text{blood pH}$ → progression to high anion gap metabolic acidosis

Diagnosis: Tachypnea and hyperpnea breathing pattern raises the suspicion of salicylate toxicity.

History of ingestion, characteristic signs/symptoms, acid-base findings, measurement of high salicylate levels

Treatment: Assess severity of the toxicity, stabilize, GI decontamination, correct metabolic abnormalities, ***serum and urine alkalinization with sodium bicarbonate***, frequent laboratory monitoring, hemodialysis if indicated.

From: 38 Anti-inflammatory, Antipyretic, and Analgesic Agents

Lippincott® Illustrated Reviews: Pharmacology, 7e, 2019

The figure illustrates the dose-dependent effects of salicylates.

TAKEAWAY:

Symptoms / Signs → Salicylate toxicity

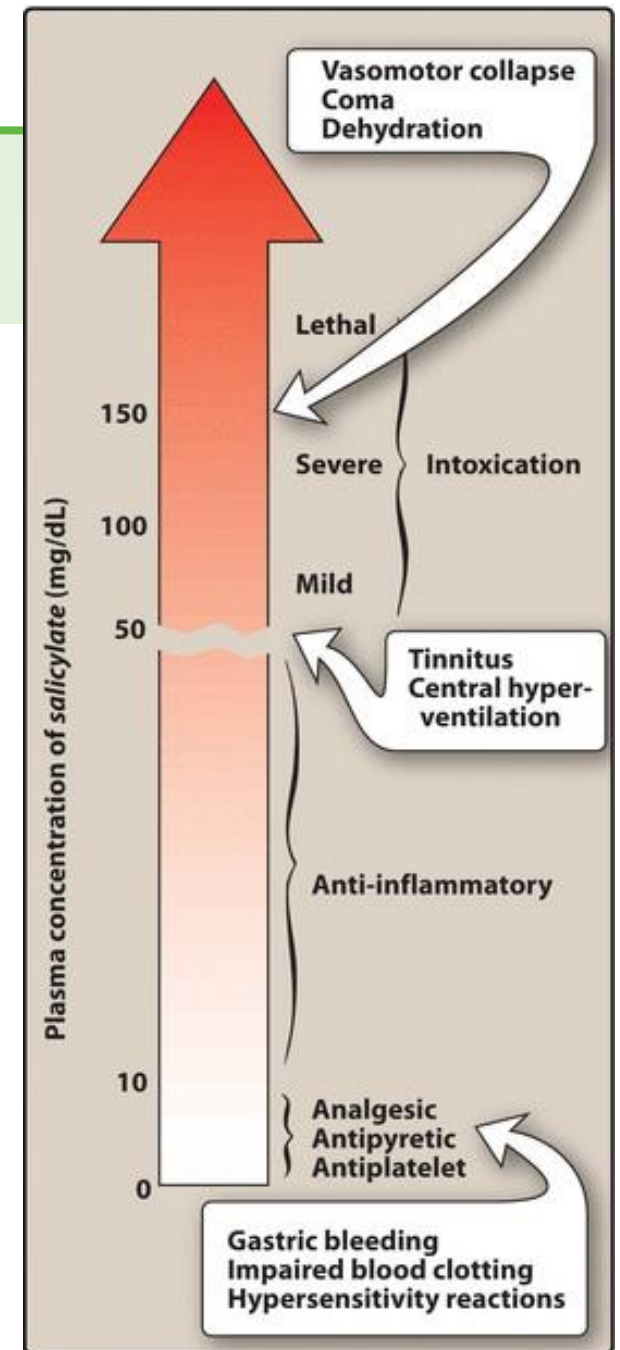
- Students should be able to identify the dose-dependent effects of acute and chronic salicylate toxicity.

Case vignette:

A patient presents with... → Recognize salicylate toxicity.

Treatment

Assess severity of the toxicity, stabilize, GI decontamination, correct metabolic abnormalities, **serum and urine alkalization with sodium bicarbonate (ion trapping mechanism)**, frequent laboratory monitoring, hemodialysis if indicated.



Contraindications, Precautions: Aspirin & Salicylate Use

Avoid in patients with:

GI disease
Coagulation disorders
Congestive heart failure
Hypovolemia
Severe hepatic
impairment
Renal impairment
Drug interactions

Contraindications

- ▶ Aspirin-, NSAID-exacerbated respiratory disease (AERD / NERD)
 1. Asthma
 2. Rhinosinusitis with nasal polyposis
 3. Acute upper / lower airway hyperreactivity causing nasal congestion and bronchoconstriction
- ▶ Viral illness in children and adolescents
- ▶ Elderly

Check your knowledge

How does the structure of aspirin relate to its mechanism of action?

What is the key principle behind aspirin's "cardioprotective" effect?

How?

Why doesn't salicylic acid inhibit platelet aggregation even though it has analgesic, anti-inflammatory, and antipyretic activity?

Check your knowledge

What is the most common adverse effect of aspirin use?

What is AERD and risk factors?

What is the proposed mechanism of AERD and its manifestations?

Can non-aspirin NSAIDs be taken by people with a history of AERD?

Check your knowledge

A 22-year-old college student presents to the university medical center ED with ringing in her ears, nausea, and rapid deep breathing. The patient reports taking “a lot of aspirin – not sure how many” for severe tension headaches during Comp week. On examination, they appears anxious and mildly diaphoretic, respiratory rate is 28 breaths/min, and pulse 110 beats per minute (tachycardia). Arterial blood gas analysis reveals a primary respiratory alkalosis with a developing high anion gap metabolic acidosis. Serum levels are consistent with drug toxicity.

History, symptoms and signs are consistent with_____.

Applying what you learned about the behavior of weak acids in solution, what is the purpose of administering intravenous bicarbonate to this patient?

Non-Acetylated (Non-Aspirin) Salicylates

Diflunisal:

Salicylic acid derivative not metabolized to salicylic acid

Salsalate:

Two salicylate molecules linked by ester bond and metabolized to salicylic acid

Sodium salicylate:

Active salicylate, does not require conversion

Choline magnesium trisalicylate:

Three salicylate molecules electrostatically associated with choline and magnesium and metabolized to salicylic acid

These products
are not available
in the US.

TAKEAWAY

These drugs are rarely used because they are less effective anti-inflammatory agents than other NSAIDs and have the potential to cause salicylate toxicities.

They are not “starred” and will not be tested in this course.

Students should be aware of them, and they may be tested on a national board exam.

Features of non-acetylated salicylates compared to aspirin

Mechanism of action:

Suppress COX-2 gene promotor activity → inhibits gene transcription and mRNA induction

Very weak COX-1 and COX-2 inhibitors

- Longer half-lives than aspirin (2-3 times daily dosing)
- Relatively slow onset of analgesic effect
- Less frequently associated with GI toxicity
- Do not affect platelet function
- Generally tolerated by adults with asthma
- Diflunisal is not converted to salicylic acid and does not cross BBB (not effective in the treatment of fever).

NON-SALICYLATE NSAIDS

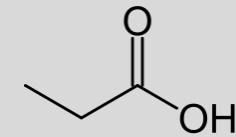
(weak acids / pKa 3-5)

Propionic Acid Derivatives

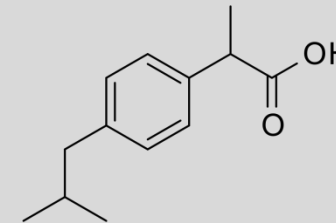
*Ibuprofen

*Naproxen

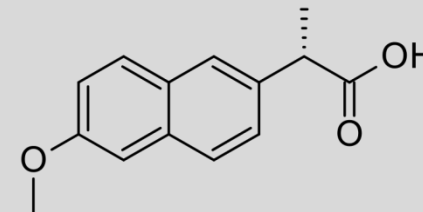
- Fenoprofen
- Ketoprofen
- Flurbiprofen
- Oxaprozin
- and others



propionic
acid



ibuprofen



naproxen

Pharmacokinetics: **TAKEAWAY** – Typical of NSAIDs class

	Ibuprofen	Naproxen
A	I.V.; Oral, rapid absorption Bioavailability ~80%	Oral, rapid absorption Bioavailability 95%
D	Protein binding >99% Widely distributed	Protein binding >99% Widely distributed
M	Hepatic metabolism (oxidation)	Hepatic metabolism (oxidation)
E	Renal excretion of metabolites	Renal excretion of metabolites
$t_{1/2}$	~2 hours; up to 50 hours in premature infants (highly variable)	12-17 hours – relatively long half-life
Onset	Analgesic 30-60 min Antipyretic ~60 min (single dose)	Analgesic 30-60 min
Duration	Antipyretic 6-8 hours (oral)	Analgesic effect ~12 hours

Ibuprofen and Naproxen:

Reversible COX-1 and COX-2 inhibitors

IBUPROFEN (OTC and Rx)

Monotherapy

Combination products:

opioids, antihistamines,
decongestants

Approved for infants, children,
and adolescents:

Closure of PDA (I.V.)

- Pain
- Fever
- Juvenile arthritis

PDA: patent ductus arteriosus

NAPROXEN (OTC and Rx)

Monotherapy

Combination products:

antihistamines, decongestants,
proton pump inhibitor, triptan

Age-specific indications in children:

- Pain, fever (≥ 12 years old - OTC)
- Juvenile arthritis (>2 yrs old)

Weight-based pediatric dosing for is listed in
the drug monographs.

Adverse effects are typical of nsNSAIDs class, including increased cardiovascular risk

Ibuprofen and naproxen **taken before low-dose aspirin** have been shown to inhibit aspirin binding to COX-1 in platelets.

Here is a suggested mechanism: *(Figure on the next slide)*

1. Ibuprofen and naproxen reversibly bind to the active site of platelet COX-1.
2. Aspirin taken after the ibuprofen or naproxen is unable to bind to the serine residue within the hydrophobic channel.
3. Aspirin is rapidly hydrolyzed before the nsNSAID dissociates from the enzyme.
4. When the ibuprofen or naproxen dissociates, platelet COX-1 is no longer inhibited.

TAKEAWAY: Low-dose aspirin should be taken before the nsNSAID.

Caution: The combination increases the risk of GI adverse effects.

Interaction between aspirin, ibuprofen, and rofecoxib, based on Catella-Lawson and colleagues.

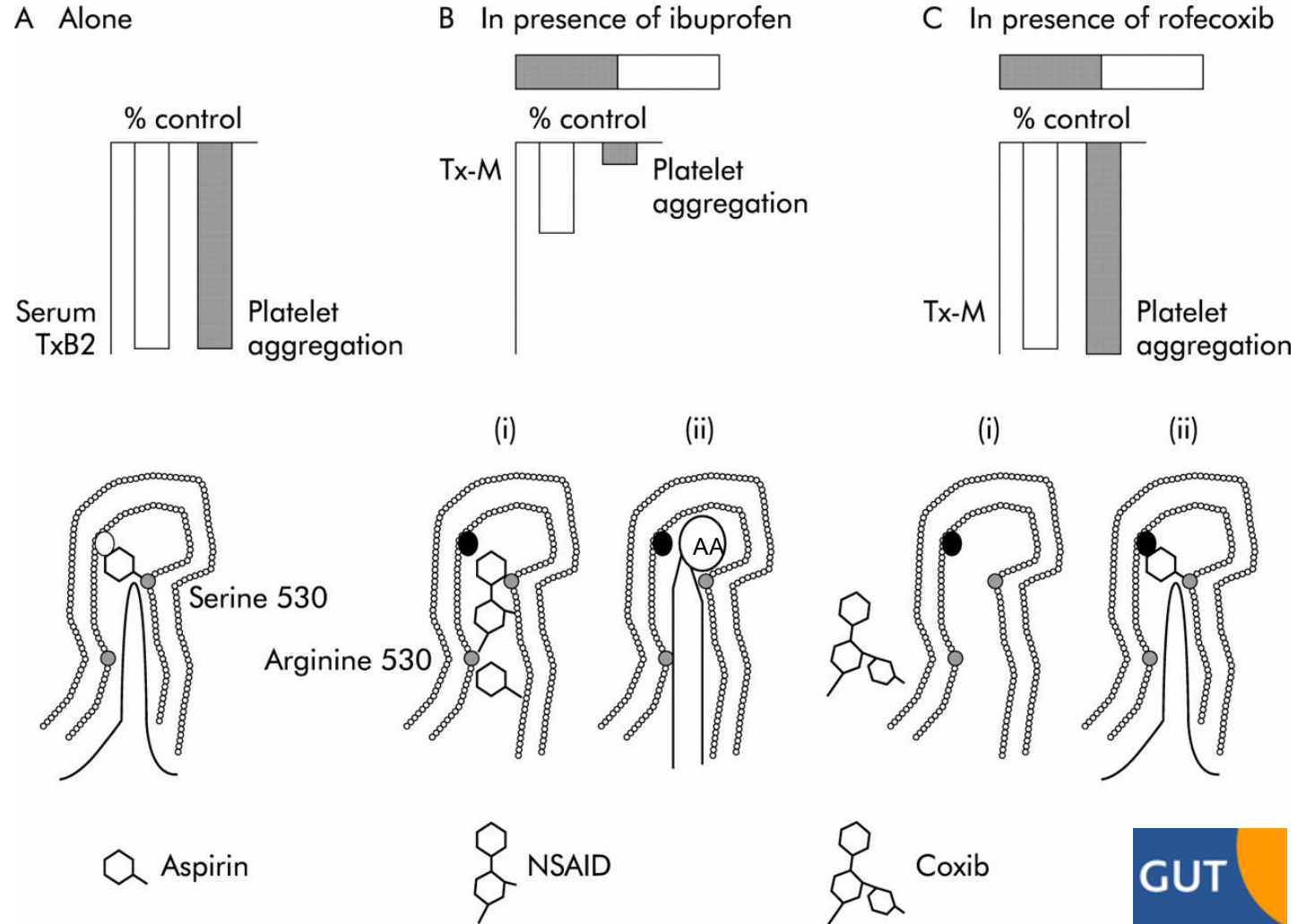
Panel A: Aspirin 81 mg reduces serum thromboxane (Tx_{B2}) by $\geq 98\%$ and platelet aggregation by 98% (at 24 hours).

TxB₂ is a biomarker of TXA₂ production.

Panel B (i): In the center, ibuprofen given 2 hours before aspirin impedes aspirin binding.

Panel B (ii): When IBU dissociates TXA₂ can be synthesized.

Panel C: The Coxib does not interact with COX-1 and therefore does not impede aspirin binding.



Hawkey C J , Langman M J S Gut 2003;52:600-608

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Check your knowledge

How would you sum up the PK properties, mechanism of action, therapeutic uses, and adverse effects of ibuprofen and naproxen in relation to the NSAIDs class?

Why is the adult dose of naproxen typically twice daily as need for pain, while ibuprofen is taken every 4 to 6 hours as need for pain (dose-dependent)?

A patient taking a daily oral low-dose aspirin for prevention of myocardial infarction also takes ibuprofen as needed for occasional muscle pain. What advise would you give about taking the two drugs concurrently?

Check your knowledge

What is the mechanistic basis for this advice?

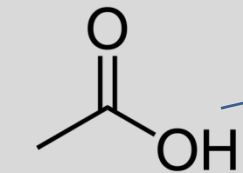
Acetic Acid Derivatives

* Indomethacin

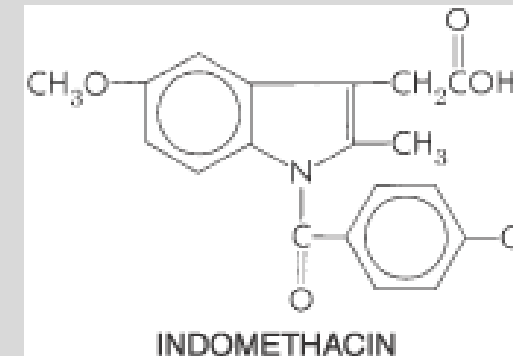
* Ketorolac

* Diclofenac

- Sulindac
- Etodolac
- Nabumetone
- Tolmetin



acetic acid



INDOMETHACIN

Pharmacokinetic properties: **TAKEAWAY – Typical of NSAIDs Class**

	Indomethacin	Ketorolac	Diclofenac
A	Oral; I.V.; rectal Rapid, complete absorption from GI tract	I.V.; I.M. Intranasal (systemic effect) Ophthalmic (topical) Oral	Oral; I.V. Transdermal Topical (acute MSK pain) Topical (actinic keratosis) Ophthalmic (topical) Rectal (not available in US)
D	Protein binding 99% Widely distributed Accumulates in synovial fluid Crosses blood-brain barrier	Protein binding >99% Poor penetration into CSF	Protein binding >99% Widely distributed Accumulates in synovial fluid Crosses blood-brain barrier
M	Hepatic; inactive metabolites minor CYP involvement Enterohepatic circulation	Hepatic	Hepatic
E	Urine, biliary (feces)	Urine, unchanged (~60%)	Urine, biliary (feces)
t _{1/2}	~2.5-3.5 hours 12-20 hours in neonate	2-9 hours	1-2 hours
		Onset 20-30 minutes Duration 4-6 h (analgesia)	

Activity: Non-selective COX inhibitors **TAKEAWAY – shaded, bold, red arrows**

	Indomethacin	Ketorolac	Diclofenac
PD	Affinity for COX-1 > COX-2	Affinity for COX-1 > COX-2	Affinity for COX-2 > COX-1
	Potent COX inhibition, (10-40x aspirin) Also: • Inhibits chemotaxis • Reduces neutrophil migration • Decreases T, B cell proliferation • Decreases inflammatory cytokines	Low dose, short duration: • Potent, opioid-like analgesic effects. • High risk of GI adverse effects compared to other NSAIDs.	COX-2 potency is similar to the COX-2 selective NSAID, celecoxib.
Uses	• Acute gout • Inflammatory and rheumatoid disorders: adults and children ≥2 years old • Closure of PDA (IV) • Tocolytic (suppresses premature labor contractions)	Excellent analgesic • Acute pain: in-hospital treatment Low dose = Not useful for Tx of systemic inflammation • Ophthalmic: keratitis, post-op inflammation and pain	Analgesia, Arthritis Acute migraine treatment Topical (derm): Joint and muscle pain Actinic keratosis (atypical epidermal keratinocytes) Ophthalmic: post-op inflammation / pain

Typical NSAIDs class adverse effects and drug interactions plus some drug-specific effects:

Indomethacin

CNS:

severe headache,
dizziness, depression

GI, common:

nausea, vomiting,
heartburn, diarrhea

***Many patients are
unable to tolerate the
side effects.***

Ketorolac

GI bleeding

- Can occur at ***any time*** during use and ***without warning***
- Can be fatal.
- ***5 days maximum*** duration of systemic ketorolac
- ***Elderly patients are at greatest risk.***

and perforation

Diclofenac

CNS effects

GI adverse effects

Cardiovascular risk

**highest CV risk of the
nsNSAIDs**

Hepatotoxicity

increased serum
aminotransferases

Rash

Photosensitivity

Check your knowledge

Which NSAID is notable for its excellent analgesic effects?

Why is this drug not used for the management of inflammation?

Check your knowledge

A 54-year-old man presents with severe pain, redness, and swelling of his right big toe. He has a history of recurrent acute gouty arthritis. The patient is offered treatment with an NSAID that is a potent inhibitor of COX-1 and COX-2 and likely suppresses immune cell function. The patient replies that last time he took the drug, it gave him “terrible headaches”, dizziness, and abdominal pain.

What is the described drug?

What two drugs are FDA approved for closure of the ductus arteriosus in neonates?

A 62-year-old person with localized knee pain due to mild osteoarthritis externally applies a topical NSAID gel to the area up to four times daily as needed. The COX affinities of the active ingredient are $\text{COX-2} > \text{COX-1}$. When taken orally, the drug is known to have a relatively high cardiovascular risk and cause elevated serum transaminase levels (liver enzymes).

What is the described drug?

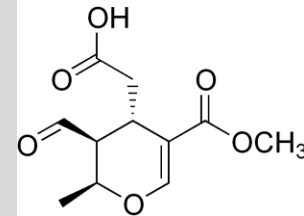
Enolic Acid Derivatives

* Meloxicam

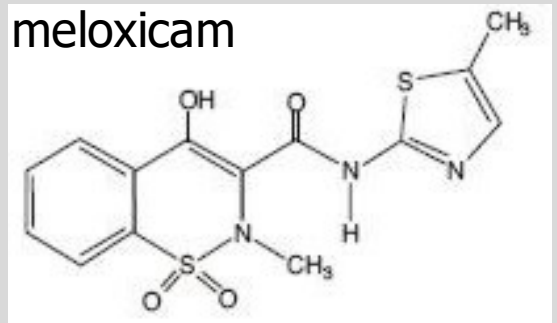
- Piroxicam

long-acting NSAIDs

enolic acid



meloxicam

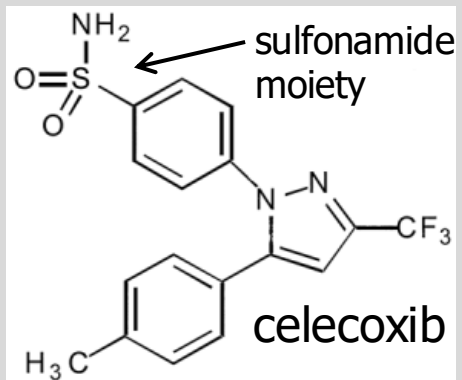


Meloxicam and Piroxicam Properties

	Meloxicam (oral, I.V.)	Piroxicam (oral)
t _{1/2}	15-24 hours (1x/day)	~50 hours (1x/day)
PD	Preferentially inhibits COX-2 Potency similar to celecoxib ➤ Selectivity decreases at higher doses	Nonselective COX-1 and COX-2 inhibitor
Uses	Mild-to-moderate pain and inflammation Osteoarthritis and rheumatoid arthritis symptoms I.V. for pain (slow onset, 2-3h)	Chronic management of rheumatoid arthritis and osteoarthritis pain and inflammation
AEs	Generally well tolerated Side effects typical of coxibs at low doses (7.5mg/d) Gastric injury incidence reduced at low dose but increased rate at higher dose (15 mg/d)	More GI and serious skin reactions observed than with other nonselective NSAIDs
DDIs	CYP2C9-mediated	CYP2C9-mediated

**Key
points**





potential for cross-reactivity with sulfonamide antibiotics?

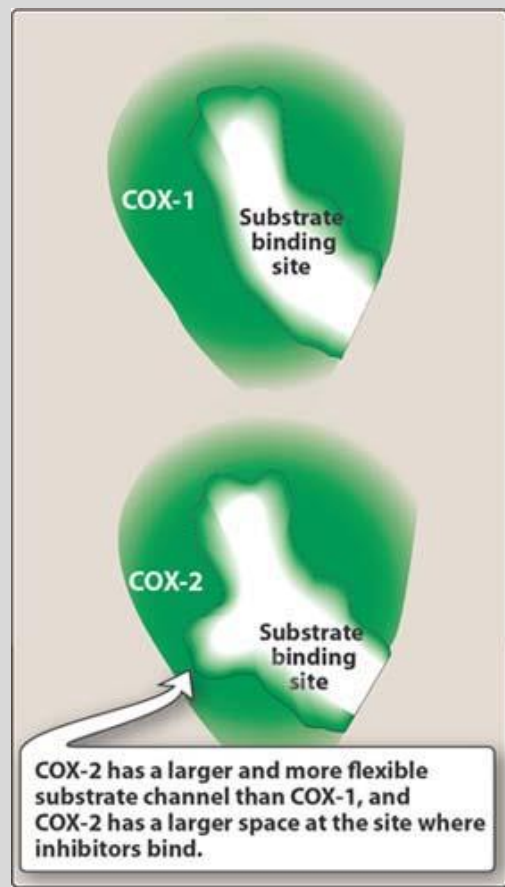
COX-2 Selective NSAIDs The “Coxibs”

* Celecoxib

Structure-activity relationship:

Coxibs' bulky structure fits in COX-2 the side pocket → steric hindrance prevents AA entry.

The molecule is too large to fit the COX-1 narrow channel.



Celecoxib Therapeutic Uses and Efficacy

PK: typical of class, $t_{1/2}$ ~11 hours (adults) → 2x daily dose

Therapeutic uses

Analgesic, Antipyretic, Anti-inflammatory

- Acute pain (adults)
- Primary dysmenorrhea
- Anti-inflammatory
 - Osteoarthritis
 - Rheumatoid arthritis
 - Ankylosing spondylitis
 - Juvenile arthritis (≥ 2 years old)

TAKEAWAY

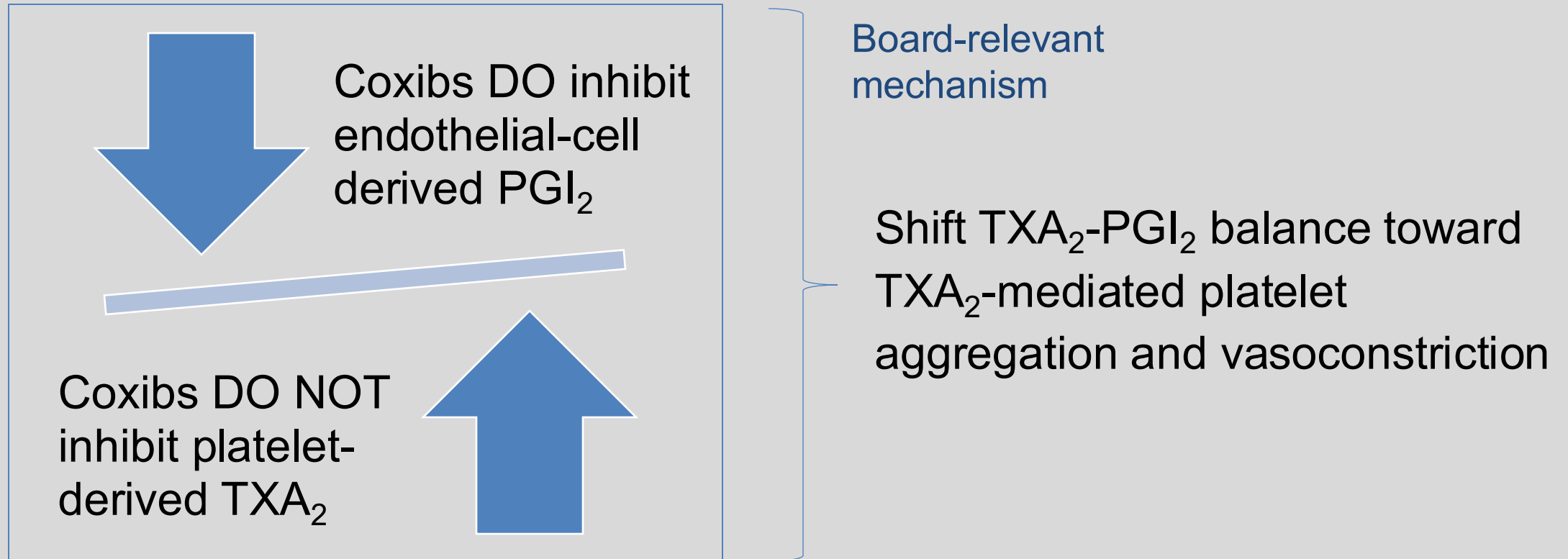
- Efficacy of celecoxib for pain and inflammation is equivalent to the nsNSAIDs
- with better GI safety and
- cardiovascular and renal risk comparable to ibuprofen and naproxen.

Adverse effects: Typical of NSAIDs class:

GI (when taken with low-dose aspirin), renal (constitutive and inducible COX-2), hypertension / edema (Na^+ /water reabsorption), avoid in CABG perioperative period (all NSAIDs), hypersensitivity reactions

Proposed mechanism of cardiovascular adverse effects associated with selective COX-2 inhibition.

Historical context: Trials with early selective COX-2 inhibitor, rofecoxib, indicated increased risk of coronary and cardiovascular death, even in small doses.



Other factors may contribute, including effects on blood pressure, renal function, endothelial cells, and nitric oxide production.

Check your knowledge

How does celecoxib compare to nsNSAIDs in terms of efficacy?

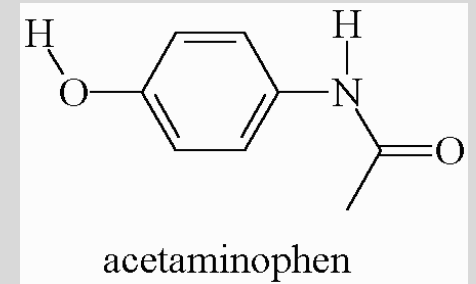
How does celecoxib compare to nsNSAIDs in terms of risk of GI adverse effects?

A 30-year-old recreational runner presents with gradually worsening pain in their left calf after increasing weekly mileage while training for a half-marathon. History and physical are consistent with muscle strain. RICE therapy is advised and a prescription for a 14-day supply of a nsNSAID (relative affinities: COX-2 > COX-1) is provided to reduce inflammation and provide short-term pain relief. The patient is directed to take one tablet daily for the shortest duration possible to relieve the inflammation and pain.

What is the described drug?

Acetaminophen

N-acetyl-para-aminophenol (APAP)



Acetaminophen Pharmacokinetics

A	Oral; rectal; intravenous Well absorbed from GI tract, rapid onset
D	Widely distributed; low plasma protein binding
M	1. Glucuronidation / sulfation (primary route; saturable) 2. CYP450 oxidation (5-15%); NAPQI is a toxic intermediate formed principally via CYP2E1 metabolism 3. Glutathione detoxification (glutathione depletion → ↑risk of hepatotoxicity)
E	Renal excretion of metabolites
t _{1/2}	2-3 hours (adults) Neonates 1-7 hours (immature UGT)
Duration	Analgesia 4-6 hours Antipyretic effect ≥ 6 hours

see next slide

Acetaminophen Metabolism

1) 80-95% of drug is metabolized by glucuronidation (main) and sulfation (secondary) to inactive metab. UDP-glucuronosyl transferase and sulfotransferase phase 2 metabolism

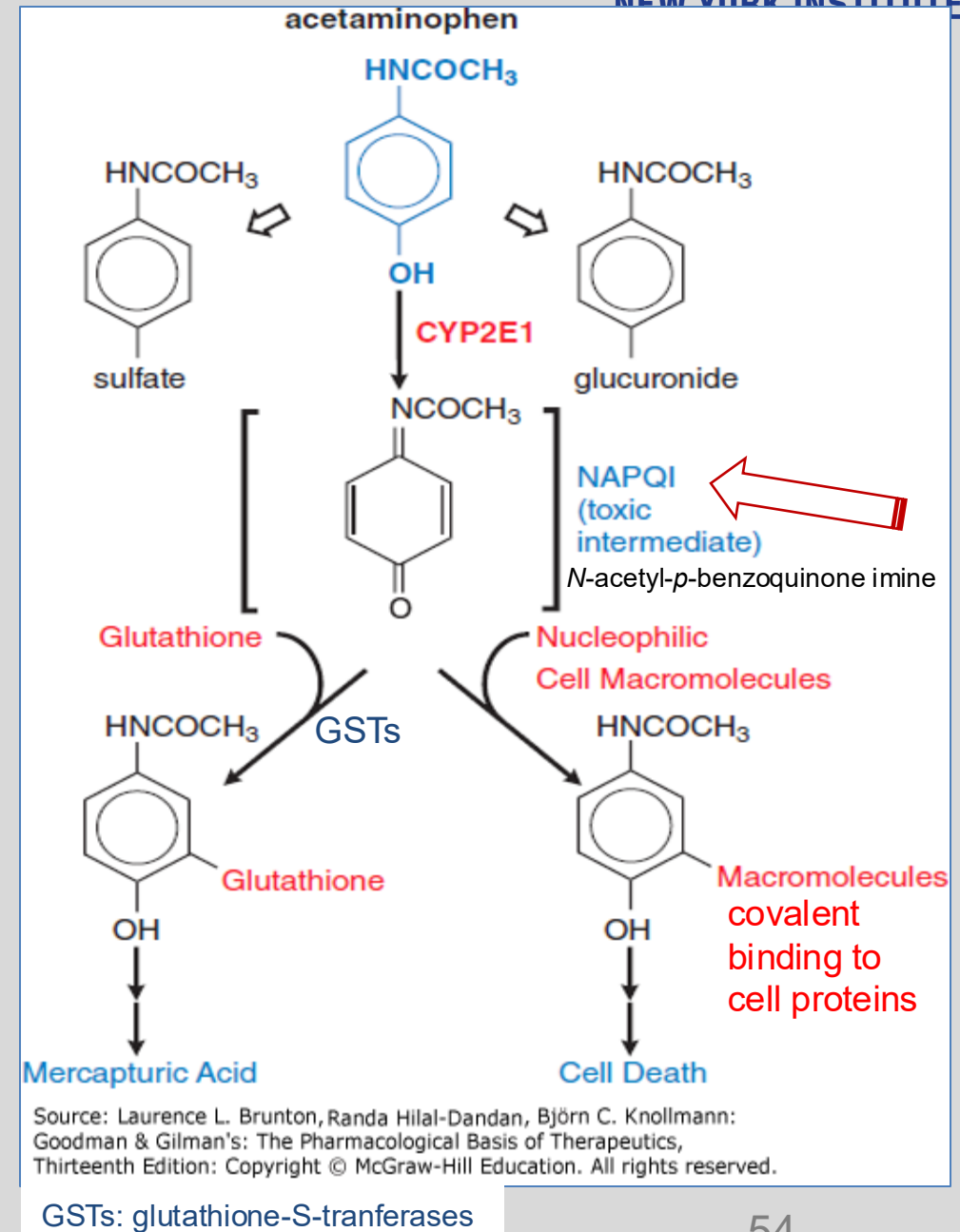
2) 5-15% P450 oxidation. CYP2E1 is the principal enzyme involved in formation of toxic NAPQI.

3) Glutathione detoxification to mercapturic acid and cysteine conjugates.

Renal excretion of conjugated metabolites (<5% unchanged drug)

Saturation of conjugation pathways and depletion of reduced glutathione can lead to NAPQI toxicity.

NAPQI binds mitochondrial proteins → oxidative stress, mitochondrial dysfunction, adduct formation → cell necrosis.



Risk factors for glutathione deficiency

Acetaminophen is a major cause of overdose and overdose-related liver failure and death.

Genetic glutathione deficiency:

Defects in synthases involved in glutathione synthesis

Acquired glutathione deficiency:

Glutathione is depleted when the body's detoxification demand exceeds production.

- Ingestion of high doses of acetaminophen.
- Nutritional deficiency

Glutathione is a tripeptide of glutamine-cysteine-glycine synthesized in the liver.

- Chronic, metabolic, and neurodegenerative diseases that increase oxidative stress and impair antioxidant production

Examples: diabetes; obesity; chronic liver disease; HIV; Alzheimer's, Parkinson's, Huntington's disease

- Older age
-

Chronic alcohol consumption with repeated supratherapeutic acetaminophen doses due to upregulation of CYP2E1 increasing production of NAPQI.

Acute alcohol ingestion does *not* appear to be a risk factor for hepatotoxicity.

N-Acetylcysteine (NAC)

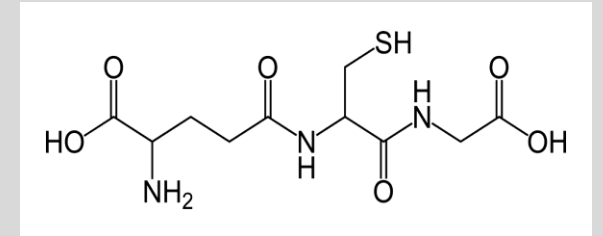
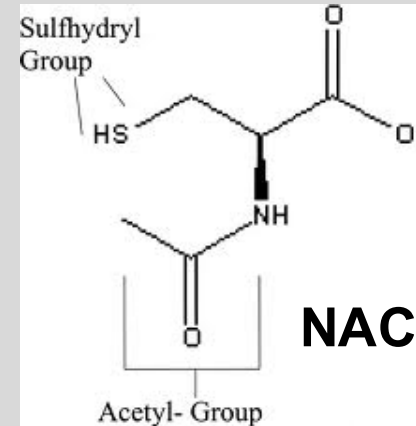
Antidote for Acetaminophen Toxicity

I.V. or oral

1. Restores hepatic glutathione
2. Serves as a glutathione substitute
3. Enhances the nontoxic sulfate conjugation of acetaminophen

Effectiveness:

Serious hepatotoxicity is uncommon and death extremely rare if NAC is administered within 8 hours following APAP overdose.



Glutathione

glutamate-cysteine-glycine tripeptide

Management of acetaminophen toxicity:

- stabilization, decontamination, NAC administration

Poisoning severity following an acute ingestion is quantified by plotting a timed serum acetaminophen poisoning nomogram. (Please refer to the toxicology resources for details.)

Acetaminophen Actions and Effects

Proposed Mechanisms of Action (not well understood)

- Inhibits COX-2 in the central nervous system
- Does not appear to bind the AA site
- Antinociceptive effects may also result from interactions with endogenous opioid, cannabinoid, and serotonergic systems.
- Lacks activity in peripheral tissues

Efficacy

- Analgesic and antipyretic effects comparable to aspirin
- **Ineffective anti-inflammatory drug**
 - High concentration of lipid peroxidase in areas of inflammation weakens acetaminophen action.
 - Acetaminophen does not inhibit neutrophil activation.

Acetaminophen Effects and Uses

Effects

- Well tolerated
- Does not appear to cause platelet dysfunction
- Low incidence of GI side effects

Therapeutic uses

- Fever
- Mild to moderate pain
- Moderate to severe pain with an opioid analgesic (allows lower dose of each)

ALERT: Use caution when prescribing, preparing, and administering I.V. acetaminophen to avoid errors that could result in accidental overdose and death.
[US Boxed Warning]

Hepatotoxicity: Mild, reversible elevated hepatic enzymes with I.V. use

Toxic doses (15 g): Severe hepatotoxicity with centrilobular necrosis

Analgesic nephropathy: Not seen with acetaminophen

Historical context: Papillary necrosis was associated with the chronic use of combined phenacetin, aspirin and caffeine. Phenacetan is metabolized to acetaminophen.

Phenacetin was marketed in 1887 and withdrawn in US in 1983. Tylenol came on the market in 1955.

Special Populations

Pregnancy:

Acetaminophen crosses placenta

Considered appropriate pain and fever treatment

Lactation: Acetaminophen is considered compatible with breastfeeding

Pediatrics: Acetaminophen is approved for use in infants, children, and adolescents.

Elderly: In older adults with hepatic impairment or history of alcohol abuse being treated for persistent pain, do not exceed a maximum of 2,000 to 3,000 mg/day (2 extra strength tablets 2 to 3 times daily).

Acetaminophen Drug Interactions

CYP inducers:

May decrease effectiveness of acetaminophen

CYP2E1 induction → ↑rate of acetaminophen metabolism → ↑ rate of NAPQI production → **may** increase risk of hepatotoxicity (see risk factors)

Warfarin: Occasional acetaminophen use is considered safe

Daily acetaminophen use (>1.3-2 g/day) may enhance anticoagulant effect.

1 adult dose = 650 mg (regular strength) every 3-4 hours as needed

1.3-2 g/day ≈ 2-5 doses per day

Mechanism not established

Cholestyramine: ↓ absorption

the resin binds acetaminophen in GI tract

Check your knowledge

Your 17-year-old cousin tells you that acetaminophen really works to relieve headaches from “studying too much”. She wonders why it didn’t help with the ankle sprain pain and swelling she got from playing soccer and ibuprofen did help.

What do you tell her?

How is the relative efficacy of acetaminophen typically described?

How does acetaminophen’s side effect profile compare with the NSAIDs’ SE profile?

Acetaminophen is considered safe to use in which populations?

Check your knowledge

Why is glutathione depletion a risk factor for acetaminophen toxicity?

Summary of the Pharmacology of NSAIDs

Summary of Aspirin

- Aspirin is rapidly converted by esterases to salicylic acid, which is a reversible inhibitor with analgesic, antipyretic, and anti-inflammatory actions. Plasma protein binding and metabolic pathways are saturable at higher therapeutic doses (zero-order kinetics).
- Aspirin's primary use is in low doses for antiplatelet effects in the secondary prevention of cardiovascular events, MI, stroke, and transient ischemic attacks. Selected patients at high risk for CV adverse events may be prescribed low-dose aspirin for primary prevention. High doses have anti-inflammatory effects comparable to other NSAIDs. Aspirin is rarely used for this purpose due to its adverse effects.
- Aspirin's active metabolite, salicylic acid, and the nonacetylated salicylates are effective anti-inflammatory compounds. They do not inhibit antiplatelet function. Salicylic acid is commercially available for topical use only.

Summary of Aspirin Toxicity

- Aspirin's principal toxicity is bleeding.
- In addition to the class toxicities, aspirin-specific adverse effects are aspirin-exacerbated respiratory disease in patients with the triad of bronchospasm, chronic rhinosinusitis, and nasal polyposis, and hypersensitivity reactions.
- Symptoms and signs of salicylate poisoning (salicylism) include tinnitus, confusion, headache, visual problems, abdominal pain, tachypnea, hyperpnea, and acid-base disorders. Ingestion of toxic doses of aspirin can be fatal to children.
- Symptoms of mild chronic salicylate toxicity can be milder, often go unrecognized or are attributed to another condition, and occur at lower serum salicylate concentrations (even within the therapeutic range).

Summary: Drug-specific features of non-aspirin NSAIDs

- The properties of non-aspirin NSAIDs reflect the general class properties.
- Ibuprofen and acetaminophen treat pain and fever in pediatric patients of any age and adults. OTC naproxen is approved for treatment of pain and fever in pediatric patients ≥ 12 years old.
- Intravenous indomethacin and ibuprofen are indicated for closure of the ductus arteriosus in premature neonates.
- All non-aspirin NSAIDs are used for their anti-inflammatory effects in inflammatory and autoimmune diseases. Pediatric dosing recommendations are available.
- NSAIDs should be used at the lowest effective dose for the shortest duration possible due to their potential dose- and duration-related adverse effects.
- Low-dose aspirin should be taken at least 30 minutes before or hours after nsNSAIDs (timing depends on clearance of the nsNSAID).

Summary: NSAIDs Cautions

NSAIDs should be avoided in patients at risk for:

- GI bleeding, including bariatric surgery: development of anastomotic ulcerations/perforations; occult bleeding exacerbates anemia
- Renal disease, uncontrolled hypertension, hypovolemia, hyperkalemia
- Thrombotic cardiovascular complications, including coronary artery bypass graft surgery (contraindication)
- Asthma: aspirin-exacerbated respiratory disease
- Pregnancy after 20 weeks gestation.

Drug-drug interactions can increase risk of adverse drug events when used concurrently with other drugs having the same effects. Metabolic and transport related interactions involve CYPs and OAT.

Summary of Acetaminophen

- Effective for mild to moderate non-inflammatory pain; may be opioid-sparing – meaning, a lower opioid dose may be adequate for pain control when combined with acetaminophen.
- Does not increase risk of serious GI complications.
- Does not alter platelet functioning.
- Can cause hepatotoxicity in chronic or acute overdose.
- Use a lower total daily dose in older adults and in patients at risk for hepatotoxicity or with organ dysfunction.

Check your knowledge

How does the structure of aspirin relate to its mechanism of action?

- Aspirin is acetylsalicylic acid. The acetyl group is covalently (irreversibly) transferred to the serine residue near the AA binding site of COX-1 and COX-2. The O-acetyl-serine within the COX hydrophobic channel creates steric hindrance that prevents AA from reaching its catalytic site.

What is the key principle behind aspirin's "cardioprotective" effect?

- Shifting the TXA₂-PGI₂ balance towards antithrombotic predominance.

How?

- Irreversibly inhibiting COX-1-derived TXA₂ in platelets, allowing unopposed COX-2-derived PGI₂ platelet disaggregation and vasodilation, and inhibition of platelet aggregation.

Why doesn't salicylic acid inhibit platelet aggregation even though it has analgesic, anti-inflammatory, and antipyretic activity?

- Salicylic acid has only weak COX-inhibiting activity and platelets are anucleate. The mechanism of action of salicylic acid is related to interfering with the promotor region of COX, which inhibits gene transcription and induction of mRNA.

Check your knowledge

What is the most common adverse effect of aspirin use?

- GI: ulcer, bleeding, perforation due to inhibition of COX-1-derived PGE2 and PGI2 cytoprotective effects.

What is AERD and risk factors?

- Non-allergic aspirin exacerbated respiratory disease
- Can occur in individuals with the triad of upper and lower respiratory hyperreactivity (bronchospasm / asthma), chronic rhinosinusitis, and nasal polyposis

What is the proposed mechanism of AERD and its manifestations?

- Inhibition of COX-1 pathway may **shunt** arachidonic acid to the lipoxygenase pathway → overproduction of leukotrienes → upper and lower airway inflammation → nasal congestion and bronchospasm

Can non-aspirin NSAIDs be taken by people with a history of AERD?

- No. Most nsNSAIDs strongly inhibit COX-1 and also trigger the same leukotriene-mediated reaction → NERD.
- Selective COX-2 inhibitors, however, have a low potential for triggering NERD.

Check your knowledge

A 22-year-old college student presents to the university medical center ED with ringing in her ears, nausea, and rapid deep breathing. The patient reports taking “a lot of aspirin – not sure how many” for severe tension headaches during Comp week. On examination, they appears anxious and mildly diaphoretic, respiratory rate is 28 breaths/min, and pulse 110 beats per minute (tachycardia). Arterial blood gas analysis reveals a primary respiratory alkalosis with a developing high anion gap metabolic acidosis. Serum levels are consistent with drug toxicity.

History, symptoms and signs are consistent with_____.

- Acute salicylic acid toxicity – acute salicylism.

Applying what you learned about the behavior of weak acids in solution, what is the purpose of administering intravenous bicarbonate to this patient?

- **Principle of Ion Trapping:** The non-ionized form of salicylic acid is lipophilic and readily crosses physiologic barriers including the CNS, while the ionized form is polar and is excreted in the urine. Administration of bicarbonate increases serum and urine pH, shifting salicylate towards the ionized form, which reduces penetration of the CNS and increases salicylate excretion in urine.

Check your knowledge

How would you sum up the PK properties, mechanism of action, therapeutic uses, and adverse effects of ibuprofen and naproxen in relation to the NSAIDs class?

- Typical class effects

Why is the adult dose of naproxen typically twice daily as need for pain, while ibuprofen is taken every 4 to 6 hours as need for pain (dose-dependent)?

- Naproxen is longer acting. Its elimination half-life is ~12 hours while ibuprofen's is ~2-3 hours.

A patient taking a daily oral low-dose aspirin for prevention of myocardial infarction also takes ibuprofen as needed for occasional muscle pain. What advise would you give about taking the two drugs concurrently?

- 1) Take the low-dose aspirin first and wait at least 30 minutes to take the ibuprofen or other nsNSAID.
- 2) Take the aspirin at least 8 hours after taking ibuprofen (~4 half-lives after taking the ibuprofen).
- Chronic or repeated ibuprofen (or other NSAID) use increases the risk of ulcers and GI bleeding. *Take the lowest dose for the shortest duration possible.*

Check your knowledge

What is the mechanistic basis for this advice?

- Aspirin produces its antiplatelet effect by irreversibly inactivating the enzyme.
- The effect lasts for the life of the platelet, which is anucleate so cannot generate replacement enzymes.
- When taken before aspirin, nsNSAIDs block aspirin's access to its site of action at the serine residue in the hydrophobic catalytic channel.
- Aspirin has a ~20-minute plasma half-life, whereas ibuprofen's effects last for hours.
- Aspirin disappears before ibuprofen dissociates from the enzyme, freeing it to produce TXA₂ and platelet aggregation.
- Aspirin and nsNSAIDs independently cause GI bleeding. The risk increases when they are used together.

Check your knowledge

Which NSAID is notable for its excellent analgesic effects?

- Ketorolac has potent analgesic activity, meaning it is highly effective at low doses.
- It is for in-hospital use by IV or IM administration, usually.

Why is this drug not used for the management of inflammation?

- Ketorolac is associated with GI bleeding that can occur at *any time* during use and *without warning* symptoms. Fatalities are reported. *Elderly patients are at greatest risk.*
- Therefore, ketorolac should be used at the lowest effective dose for the shortest duration possible – **5 days maximum.**
- *Much higher ketorolac doses are needed for anti-inflammatory effect → serious adverse effects.*

Check your knowledge

A 54-year-old man presenting with severe pain, redness, and swelling of his right big toe. He has a history of recurrent acute gouty arthritis. The patient is offered treatment with an NSAID that is a potent inhibitor of COX-1 and COX-2 and likely suppresses immune cell function. The patient replies that last time he took the drug, it gave him “terrible headaches”, dizziness, and abdominal pain.

What is the described drug?

- Indomethacin

What two drugs are FDA approved for closure of the ductus arteriosus in neonates?

- Indomethacin and ibuprofen – intravenous administration

A 62-year-old person with localized knee pain due to mild osteoarthritis externally applies a topical NSAID gel to the area up to four times daily as needed. The COX affinities of the active ingredient are $\text{COX-2} > \text{COX-1}$. When taken orally, the drug is known to have a relatively high cardiovascular risk and cause elevated serum transaminase levels (liver enzymes).

What is the described drug?

- Diclofenac

Check your knowledge

How does celecoxib compare to nsNSAIDs in terms of efficacy?

- Celecoxib and nsNSAIDs are similarly effective with similar CV and renal risk.

How does celecoxib compare to nsNSAIDs in terms of risk of GI adverse effects?

- Celecoxib has a reduced risk of gastroduodenal adverse effects compared to nsNSAIDs – except when taken with daily low-dose aspirin.

A 30-year-old recreational runner presents with gradually worsening pain in their left calf after increasing weekly mileage while training for a half-marathon. History and physical are consistent with muscle strain. RICE therapy is advised and a prescription for a 14-day supply of a nsNSAID (relative affinities: COX-2 > COX-1) is provided to reduce inflammation and provide short-term pain relief. The patient is directed to take one tablet daily for the shortest duration possible to relieve the inflammation and pain.

What is the described drug?

- Meloxicam

Check your knowledge

Your 17-year-old cousin tells you that acetaminophen really works to relieve headaches from “studying too much”. She wonders why it didn’t help with the ankle sprain pain and swelling she got from playing soccer and ibuprofen did help.

What do you tell her?

- Acetaminophen works in the brain and spinal cord to block the creation of the signaling molecules, called prostaglandins, involved in pain (and fever too). It does not work well in the rest of body, especially inflamed tissues, because it does not effectively block the chemical pathways that drive swelling. Ibuprofen works differently. It blocks prostaglandins everywhere in the body, including inflamed injured or swollen areas.

How is the relative efficacy of acetaminophen typically described?

- Analgesic and antipyretic effects are comparable to aspirin.

How does acetaminophen’s side effect profile compare with the NSAIDs’ SE profile?

- Acetaminophen does not cause the serious toxicities associated with NSAID use and has few drug interactions.

Acetaminophen is considered safe to use in which populations?

- Everyone who doesn’t have a contraindication to the drug

Infants, children, adolescents, adults, elderly, pregnancy, and breastfeeding

Check your knowledge

Why is glutathione depletion a risk factor for acetaminophen toxicity?

- Acetaminophen undergoes saturable metabolism via glucuronidation, sulfation, and oxidation via CYPs.
- CYP2E1 generates NAPQI, an unstable toxic intermediate.
- NAPQI is rapidly detoxified by glutathione-S-transferase → nontoxic water-soluble metabolite.
- If glutathione levels are reduced because of genetic or acquired deficiency, NAPQI may accumulate → oxidative stress, mitochondrial dysfunction, adduct formation → hepatocyte necrosis.



Ah, pain relief!

<https://www.istockphoto.com/illustrations/relief-from-pain>

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